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using the JCA model**

**LUCAS FERNANDO COSER
DIEGO AZEVEDO SIVIERO
JOSE R. DE F. ARRUDA
BELISARIO NINA HUALLPA
ANDRE DE OLIVEIRA**

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Numerical and Experimental Studies of a Poroelastic Material using the JCA Model

L. F. Cóser¹, A. Oliveira¹, B. N. Hualpa², J. R. F. Arruda², D. A. Siviero³

¹Siemens Industry Software Ltda. - São Caetano do Sul, SP, Brazil

²State University of Campinas - Faculty of Mechanical Engineering - Department of Computational Mechanics - Campinas, SP, Brazil

³São Paulo Federal Institute of Education, Science and Technology - Department of Industrial Automation - Guarulhos, SP, Brazil

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Abstract

Poroelastic materials are commonly used in passive noise control due to their low cost and relative efficient acoustic absorption characteristics in the middle to high frequency range. They are constituted by a rigid and a fluid phase responsible for the dissipation mechanisms attenuating the propagation of acoustic waves inside the material. The understanding of these phenomena and their translation into parameters existing in the mathematical formulations for poroelastic materials are of paramount importance in the design of optimized structures for choosing the proper materials for each application. This work presents studies on the validation of a melamine foam characteristics using the equivalent fluid model from Johnson-Champoux-Allard (JCA) and a rigid structure. Experiments and finite element simulations of an impedance tube using the LMS Virtual.Lab® software were executed and validated to determine the necessary material properties for the equations involved in the numerical analysis. Results show a good approximation between numerical and experimental data, which allows a better understanding on how to use the JCA model in complex simulations using poroelastic material layers for acoustic absorption purposes.

Introduction

Noise reduction has become a very important topic in product development and in many situations the engineer must make decisions without changing the current design stage to meet specified sound quality targets. Because of their low cost and availability, materials such as natural and synthetic fibers, glass wool, felts and foams are employed in such scenarios, hence the understanding of their properties is a key to deliver optimized sound absorption results. These materials can be classified as poroelastic, i.e., they are constituted by coupled solid and fluid phases in which the former is characterized by a network of pores filled by the latter [1]. During the propagation of a sound wave inside the material, three forms of vibroacoustic energy dissipation take place [2]:

- Viscous: fluid shearing over the skeleton pores;
- Thermal: irreversible heat exchanges between the phases;
- Structural: damping in the structural movement.

For many materials such as polymeric foams, the contribution of the structural damping can be neglected and the frame can be modeled as rigid. A comprehensive discussion about this topic can be found in [1] and the above dissipation mechanisms are subject of research since the beginning of the last century [3]. Although in constant evolution, they share a same common principle [3]: by means of measurable properties representative of the microscopic material internals, a macroscopic translation by means of an equivalent fluid can be determined in terms of two fundamental acoustic properties: the density and the adiabatic bulk modulus.

The main objective in the determination of the previous properties is their use in further analytical and numerical investigations by means of poroelastic material models such as the Johnson-Champoux-Allard (JCA) [4], which is broadly implemented and can represent well the material behavior in the middle to high frequency range. This is especially important in simulations, where they can be used as material layers to yield an impedance curve representing their frequency dependent characteristics. Hence, numerical analysis of the same system using different acoustic absorption materials can be tested and their effects further analyzed to provide more appropriated solutions according to specific acoustic needs.

A key aspect concerning poroelastic materials is their characterization: how to obtain all the properties used in the JCA model? Traditionally, the absorption coefficient measured by means of an impedance tube is used as the fundamental parameter in a selection scenario. Nonetheless, it can be misleading because it ignores the influence of the boundary conditions [5] and the visco-thermal phenomena happening during the propagation of a sound wave through the material. In practice there are standard experiments to directly determine all the parameters needed in the JCA model, but they are costly and not straightforward to perform due to the need of dedicated hardware and execution procedures. Another indirect approaches proposed by [6] and [7] use a modified impedance tube to yield the equivalent density and adiabatic bulk modulus curves from the JCA model, which can then be numerically fit to yield the properties sought. However, the fitting process is not straightforward [8] and is still a subject of research.

In order to verify the measurement method proposed by [7], a previously identified melamine foam sample supplied by [9] was investigated both experimentally and numerically. The goal was not

only to check if the constructed impedance tube was properly operating, but also to understand how to work with the JCA properties in a simulation scenario to achieve the same results obtained in practice. The validation of the methodology proposed by [7] was accomplished and revealed important information regarding the air cavity layer behind the material sample used in the simulation, thus confirming what was mentioned about absorption curves: although important as a starting point in material selection, they do not provide a full understanding of its inner working and may limit or mislead important choices in product design. Finally, a better understanding on how to properly work with poroelastic materials in simulations was carried out, which gives better insight for the application of such advanced structures in simulations to predict the material behavior in earlier stages of product design and help the understanding of their contribution for noise reduction needs.

The JCA model

Works on the Johnson-Champoux-Allard (JCA) [4] started with the proposal of a semi-phenomenological model to describe the complex density of an acoustical porous material with a motionless skeleton having arbitrary pore shapes. They were extended by [10] to compute the dynamic bulk modulus for the same kind of porous material by including inertial effects in the formulation. It has been broadly used among all available poroelastic materials models due to its good agreement with experimental data in the middle to high frequency range.

Model equations

The JCA equation for the equivalent density ρ_{eq} is [4]:

$$\rho_{eq}(\omega, \sigma_R, \phi_p, \alpha_\infty, \Lambda) = \frac{\rho_0 \alpha_\infty}{\phi_p} \left[1 + \frac{\phi \sigma_R}{j \omega \rho_0 \alpha_\infty} \left(1 + j \frac{4 \omega \rho_0 \eta \alpha_\infty^2}{\sigma_R^2 \phi_p^2 \Lambda^2} \right)^{1/2} \right] \quad (1)$$

where σ_R is the flow resistivity, ϕ_p the open porosity, α_∞ the tortuosity, Λ the viscous characteristic length, ρ_0 the fluid medium density, j the imaginary number and η the kinematic viscosity. The equivalent adiabatic bulk modulus K_{eq} is given by [10]:

$$K_{eq}(\omega, \phi_p, \Lambda') = \frac{\gamma P_0 / \phi_p}{\gamma - (\gamma - 1) \left[1 + \left(\frac{8 \eta}{j \omega B^2 \Lambda'^2 \rho_0} \right) \left(1 + j \frac{\omega B^2 \rho_0 \Lambda'^2}{16 \eta} \right)^{1/2} \right]^{-1}} \quad (2)$$

with Λ' the thermal characteristic length, γ the specific heats ratio and B the Prandtl number. It is worth noting that, disregarding frequency, equations (1) and (2) depend on four and two variables respectively, which will be better detailed in the next section.

Characteristic parameters

They key parameters to the JCA model are presented below. They translate the microscopic properties into measurable quantities which can model the macroscopic behavior of the material:

- Flow resistivity (σ_R): represents the impediment given to a laminar flow through the sample and has units of Pa.s/m². It has standardized measurement procedures detailed in ISO 9053 and ASTM 522-80;
- Open porosity (ϕ_p): obtained from the ratio between the volumes of the open pores through which the acoustic wave propagates and the sample volume. It can be measured with the mass difference between the dry and saturated conditions of the sample after being immersed in a well-known (water, acetone or alcohol) fluid volume;
- Tortuosity (α_∞): it is a measure of the internal deviation inside the pores compared to the direction of propagation of the sound wave and their geometrical non-uniformities. It can be also understood as a measurement of the acoustic wave path complexity inside the material;
- Viscous characteristic length (Λ): defined by the division between the volume and the surface area of the pores weighted by the squares of the velocities at their centers and walls for the limiting case when $\omega \rightarrow \infty$, where ω is the frequency. It relates the dissipation from viscous into thermal energy among the interconnecting pores and has units of meters;
- Thermal characteristic length (Λ'): has the same units as Λ and is computed by the ratio between the pore volume and its surface area. Since no weighting is applied, it is expected that the parameter will be always greater or equal to the viscous characteristic length. Also, the parameter expresses the dissipation caused by thermal exchanges in the pores of greater volume and larger superficial area.

Although all previous parameters can be measured, they usually require both sophisticated laboratory equipment and measuring techniques. Another possible approach is to experimentally determine the equivalent fluid density and adiabatic bulk modulus curves, which can be fit into the mathematical relations given by equations (1) and (2). However, as presented in [11], the inverse characterization can be a complex task to perform depending on the algorithms implemented, and extreme care must be taken in order to avoid obtaining unrealistic or incorrect material parameters.

Experiments

This section describes the steps taken in order to determine the experimental curves for ρ_{eq} and K_{eq} from a melamine sample with 40 mm thickness.

Determination of ρ_{eq} and K_{eq}

As proposed by [7], the curves for ρ_{eq} and K_{eq} can be determined by means of an impedance tube with an additional microphone at position 3 depicted in Figure 1.

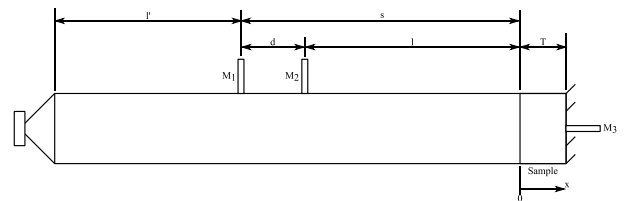


Figure 1. Generic impedance tube with an additional microphone at position 3 for the method proposed by [7]. The boundary condition behind the sample is rigid (zero velocity).

Based on the same calculations used in the standard impedance tube with a rigid wall and Figure 1, the complex reflection coefficient α_R can be calculated as [12]:

$$\alpha_R = e^{jks} \frac{H_{2,1} - e^{-jkd}}{e^{jkd} - H_{2,1}} \quad (3)$$

where k is the main fluid medium wave number, s the distance between the microphone 1 and the sample, d the distance between the microphones and $H_{2,1}$ the Frequency Response Function (FRF) between microphones 2 (response) and 1 (excitation). The material surface impedance Z_n can be found to be [12]:

$$Z_n = \rho_0 c \frac{1 + \alpha_R}{1 - \alpha_R} \quad (4)$$

with c the speed of sound in the fluid medium. Using the method proposed by [7] and referring to Figure 1, the FRF between point 0 and microphone position 3 is:

$$H_{0,3} = \frac{1 + \alpha_R}{e^{jkl} + \alpha_R e^{-jkl}} H_{2,3} \quad (5)$$

where l is the distance between the microphone 2 and the sample and $H_{2,3}$ the FRF between microphone positions 2 (response) and 3 (excitation). The material wave number k_c is then computed as:

$$k_c = \frac{1}{T} \cos^{-1}(H_{0,3}) \quad (6)$$

with T the sample thickness. The impedance at one pore, Z_{cp} , is:

$$Z_{cp} = jZ_n \tan(k_c T) \quad (7)$$

and the material characteristic impedance Z_c is [3]:

$$Z_c = \frac{Z_{cp}}{\phi_p} \quad (8)$$

Finally, the equivalent density and adiabatic bulk modulus will be [7]:

$$\rho_{eq} = \frac{Z_c k_c}{\omega} \quad (9)$$

$$K_{eq} = \omega \frac{Z_c}{k_c} \quad (10)$$

Hence, the previous equations show that experiments can be performed with the impedance tube in order to determine the equivalent fluid properties and just a few measurements with the microphones. Although not necessary neither for the calculations nor for the JCA model, the absorption coefficient α is often used as a reference parameter for acoustic absorption materials. It is calculated from the complex reflection coefficient α_R and is equal to:

$$\alpha = 1 - |\alpha_R|^2 \quad (11)$$

whose values must range from zero (total absorption) to one (total reflection).

Experimental details

Since the mathematical framework for obtaining ρ_{eq} and K_{eq} from FRFs has been presented in the previous section, the experimental execution details can now be discussed. Figure 2 shows the geometrical characteristics used in the experiments.

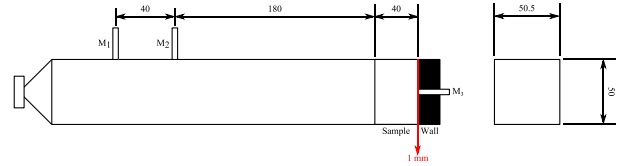


Figure 2. Impedance tube details used for both experiments and simulations. A 1 mm air cavity is considered to exist behind the sample.

As already demonstrated, all the data processing relies on the determination of FRFs between specific microphone positions, namely $H_{2,1}$ and $H_{2,3}$. In order to obtain them, a white noise ranging from 4 to 2048 Hz was applied to the speaker shown in Figure 2 so the H_1 estimator could be computed with:

$$H_1(\omega) = \frac{S_x S_y^*}{S_x S_x^*} \quad (12)$$

where S is the signal spectrum, the subscripts x and y refer to the input and the output of a linear system and the star symbol denotes a complex conjugated quantity. Since white noise has been used as the excitation signal, it is necessary to compute as much averages as possible to determine the FRF with the estimator from equation (12). Also, a Hanning window with 66% overlap has been applied to improve the results without data loss due to the windowing process.

One important remark regarding the experiments refers to the phase mismatch between the microphones: if two independent microphones are to be used, then a calibration function H_c would be necessary as detailed in [13] to correct the phase difference between them. However, this increases experimental complexity and was avoided by noting that a single microphone could be used by referring all measurement positions to a same signal. This approach yields:

$$H_{i,gen}(\omega) = \frac{S_i S_{gen}^*}{S_{gen} S_{gen}^*} \quad (13)$$

where the indexes i and gen refer to the i^{th} response position and the generator signal, respectively. Therefore, once the data from equation (13) is available for all microphone positions, $H_{2,1}$ and $H_{2,3}$ can be computed as FRF divisions given by:

$$H_{2,1}(\omega) = \frac{H_{2,gen}}{H_{1,gen}} \quad (14)$$

$$H_{2,3}(\omega) = \frac{H_{2,gen}}{H_{3,gen}} \quad (15)$$

thus yielding the necessary dataset for the calculation of ρ_{eq} and K_{eq} .

Simulations

The simulations were done using the Finite Element Method (FEM) acoustic module available in LMS Virtual.Lab® to reproduce the executed measurement with the impedance tube and the melamine test sample. The mesh was constituted by a continuous prismatic geometry with three main fluid sections as demonstrated in Figure 3: the air between the tube input and the sample wall, the equivalent fluid given by the JCA properties representing the test specimen and the air in the cavity behind the sample.

The same experimental conditions were reproduced and are detailed in Table 1. The JCA poroelastic material properties were taken from [9] and are summarized in Table 2, and it is important to remark that the sample used in the experiments belong to the same set used by [9] for the material characterization. Regarding the boundary conditions, all faces are considered to be rigid and therefore have zero velocity, with the exception of the input where a unitary surface velocity for all frequencies ranging from 4 to 2048 Hz has been applied to. Pressure spectra were taken at the microphone locations shown in Figure 3 reproducing the locations from Figure 1, with further geometrical details are given in Figure 2.

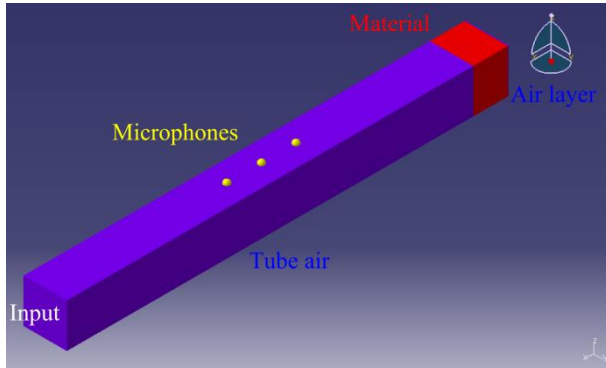


Figure 3. Impedance tube model mesh generated in LMS Virtual.Lab® showing the following element sections: tube air, poroelastic material and air layer behind the sample. The microphone positions also follow the ones found in the physical device.

It is worth to say that all processing follows the same steps used in the experimental case to compute the simulated curves for ρ_{eq} and K_{eq} . However, since the frequency spectra at the microphone positions are known and noise is absent, no estimators are necessary and the FRFs can be computed simply as:

$$H_{ij}(\omega) = \frac{S_i}{S_j} \quad (16)$$

where i and j correspond to the response and the excitation position numbers respectively.

Table 1. Environmental conditions used in the simulations.

Property name	Symbol	Unit	Value
Specific heats ratio	γ	-	1.4
Ideal gas constant for dry air	R	J/kg/K	287.058
Prandtl number	B	-	0.6861
Kinematic viscosity	η	N.s/m ²	1.782.10 ⁻⁵
Ambient temperature	T_0	°C	25.5
Ambient pressure	P_0	Pa	94700
Speed of sound	c_0	m/s	346.442
Density	ρ_0	kg/m ³	1.105

Table 2. Poroelastic material properties used for the melamine foam taken from [9].

Property name	Symbol	Unit	Value
Flow resistivity	σ_R	Pa.s/m ²	12450
Open porosity	ϕ_P	-	0,85
Tortuosity	α_∞	-	1,05
Viscous characteristic length	A	m	123.10 ⁻⁶
Thermal characteristic length	A'	m	159.10 ⁻⁶

A very important aspect in the simulation is the inclusion of the air layer behind the sample: it is impossible to guarantee a rigid backing in the experiment, and a small cavity must be included in the simulation to capture the phenomena happening in this region. Although not shown, mesh refinement does not play a role in the overall results, and different cavity thicknesses also did not present significant changes in the data obtained even when extended up to 10 mm. However, for such thickness values and above the behavior of the system will certainly change, thus requiring modifications in the equations of the method proposed by [7]. Nonetheless, this new set of equations can lead to better experimental results since now the error introduced by the sample positioning before the rigid wall is diluted into a much bigger region already embedded in the formulation.

Results

Figure 4 demonstrates the acoustic absorption coefficient computed using equation (11). The following observations can be made:

- Both simulation datasets provide similar absorption curves, i.e., apparently there is no influence from the air cavity in the results;
- Experimental data is not valid at low and high frequencies due to problems with the excitation speaker and imposed test frequency range limitation up to 2048 Hz respectively;
- Compared to the simulation data, the experimental case presents higher acoustic absorption at lower frequencies. The opposite can be observed at higher frequencies;
- There is an inflection around 620 Hz which can be explained by a structural resonance from the impedance tube at this frequency.

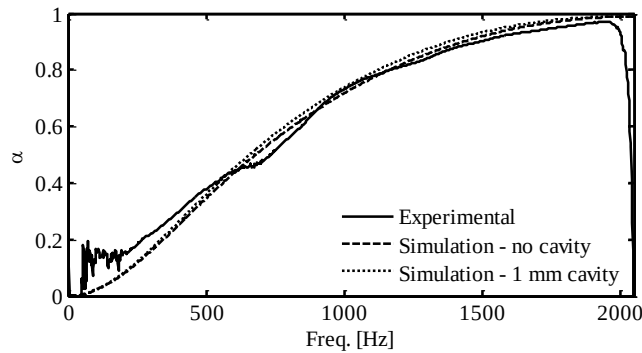


Figure 4. Absorption coefficient obtained from the experiment and the simulations without and with 1 mm air cavity behind the sample.

Figure 5 presents the real and imaginary parts of ρ_{eq} for the following cases: experimental, computed JCA model using equation (1) and simulations without and with 1 mm cavity. A very good correlation can be seen between the simulation and the JCA model, and even though with an offset and some discrepancies in the real part, the experimental curves follow the same trend observed in the previous ones. By contrast, the results from the simulation without the air cavity totally disagree with the other curves, therefore clearly confirming the necessity of its inclusion in the simulation.

For Figure 6 containing the real and imaginary parts of K_{eq} , the same observations from Figure 5 can be made. The absence of the air cavity causes worsening effects in the imaginary part of the results, whose values cannot be even shown due to major unrealistic amplifications.

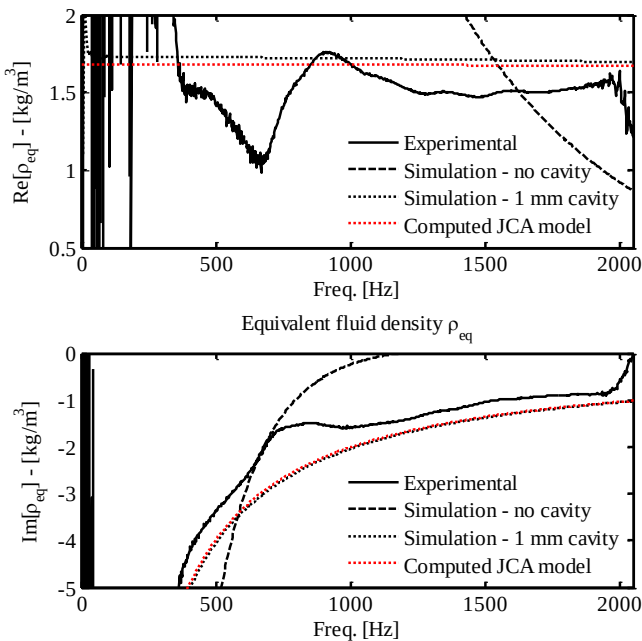


Figure 5. Comparison showing the real and imaginary parts of the equivalent fluid density for the following cases: experimental, simulation without and with 1 mm air cavity and computed JCA model using equation (1).

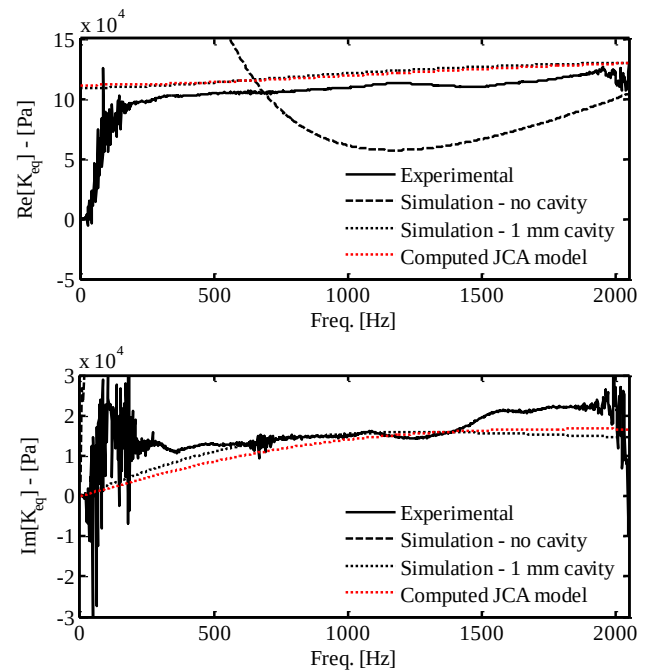


Figure 6. Comparison showing the real and imaginary parts of the equivalent fluid adiabatic bulk modulus for the following cases: experimental, simulation without and with 1 mm air cavity and computed JCA model using (2).

Conclusions

The method proposed by [7] was investigated and verified by experiments and simulations for a melamine foam with known poroelastic properties identified by [9]. Results were presented in terms of the curves for the absorption coefficient and the equivalent density and adiabatic bulk modulus of the JCA model. The former is very similar in all cases and the latter presents strong divergences in the simulation data when the air cavity behind the sample is neglected, which must be included in the simulation to correctly reproduce the experimental condition. This fact demonstrates that the information contained in the absorption curves is very limited and cannot capture all the details necessary to describe the investigated experiment. As expected, once the parameters are known, the JCA model describes well the behavior of the poroelastic material in the middle to high frequency range, therefore enabling its utilization in more advanced studies requiring acoustic absorption layers with poroelastic materials.

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Contact Information

Siemens Industry Software Ltda.

Lucas F. Cóser - lucas.coser@siemens.com

André Oliveira - andre.oliveira1@siemens.com

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Definitions/Abbreviations

JCA	Johnson-Champoux-Allard model
FEM	Finite Element Method
FRF	Frequency Response Function